African and Indigenous American ancestry shape plasma CNS biomarker baselines within and across populations, modifying Alzheimer's disease trajectories

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THE QUESTION

A biomarker reads differently in two populations: disease, ancestry, or recruitment site? Does it change the test?

THE ANSWER

Ancestry moves the baseline

- 62 of 126 proteins differ; African vs HI-WH 49, all the same sign in both lots
- a known variant accounts for CHIT1 (78%) and part of IL6R (45%)

Ancestry rarely moves the AD signal

- p-tau rises with AD in every group, in both lots
- one BD-pTau-217 cut: same share of normals above it, fewer clinical AD at most African sites

Few regional differences survive site

- Peruvian and Ecuadorian vs Caribbean 10; East vs West Africa 5
- 6 of the 15 immune

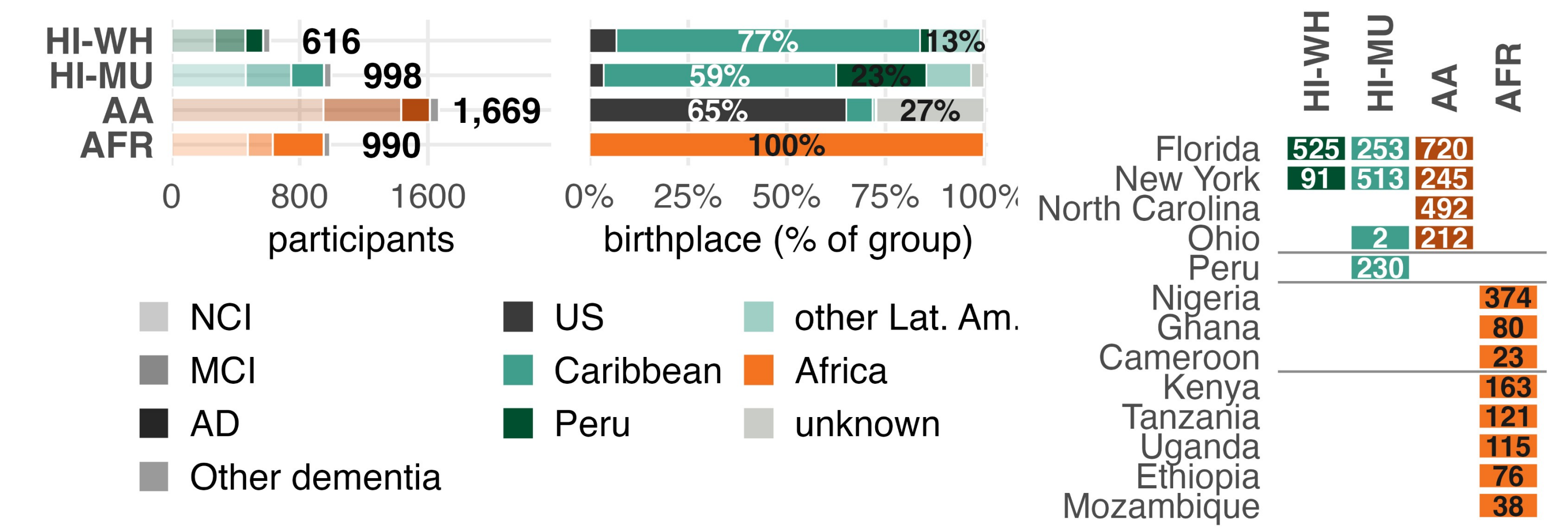
WHO AND HOW

4,273 participants

22 sites

50 countries of birth

126 plasma proteins



female 72 / 74 / 79 / 56%; median age 73 / 71 / 71 / 72 (HI-WH / HI-MU / AA / AFR)

HI-WH Hispanic, European descent · 3 sites
HI-MU Hispanic, mixed origin · 6 sites
AA African American · 6 sites
AFR African · 13 sites
22 sites in all: five serve more than one group

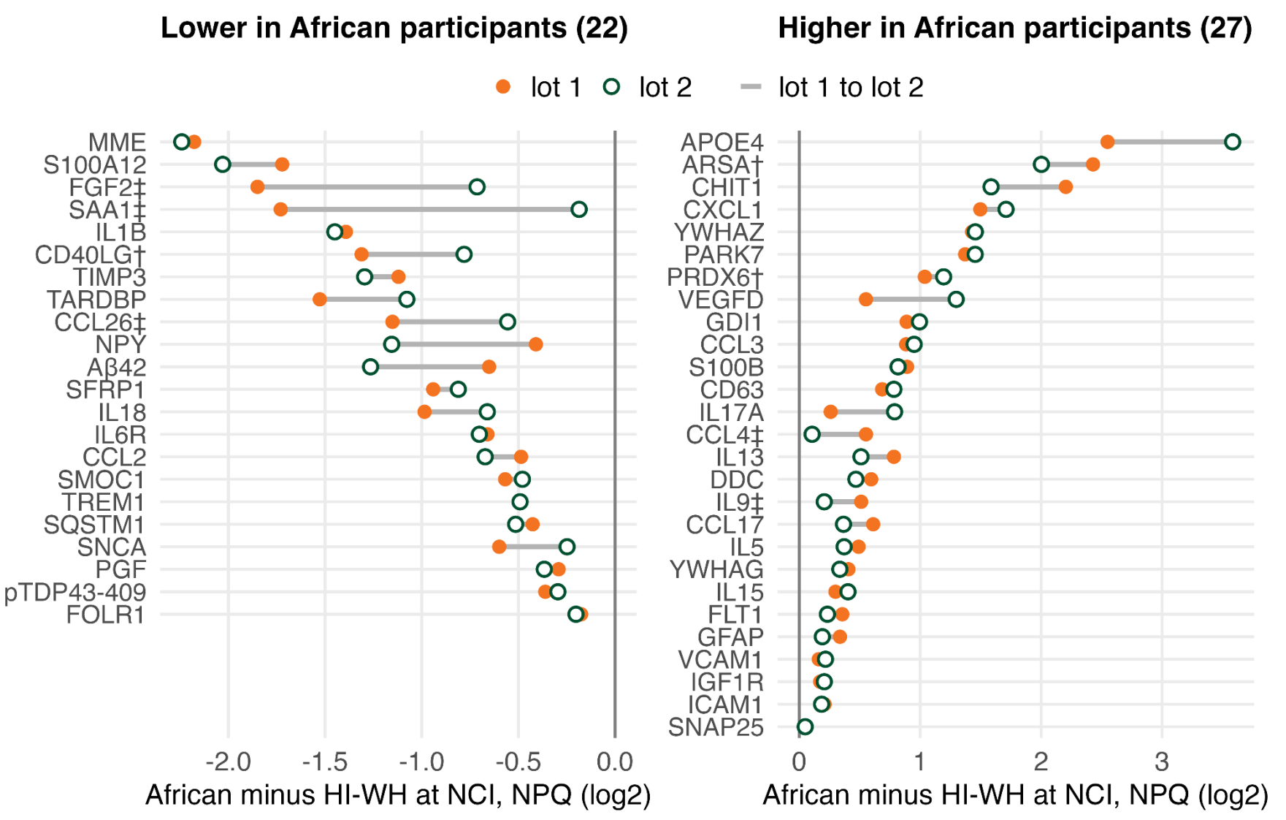
- Alamar NULISAseq CNS panel: 131 targets, 126 proteins after the hemolysis filter
- NPQ: NULISA Protein Quantification, a log2 relative level
- 2 reagent lots, each fitted alone: lot 1, 2,467 participants (2025); lot 2, 1,806 (2026)
- model: ancestry, sex, age, diagnosis, reads and a blood-cell score, with site and assay run as random effects; ancestry p-values by parametric bootstrap; full methods in the supplement

APOE ε4: THE COUNTER-EXAMPLE

- among the cognitively normal, no evidence that the ε4 effect on p-tau differs by group (pTau-217 p 0.75 in lot 1, 0.55 in lot 2)
- the group difference in the ε4 slope is a lot-1 result that comes with the diagnosis mix (clinical AD: African 38%, AA 9% of those fits)
- ε4 dose at NCI 0.22 / 0.20 / 0.44 / 0.49; figure in the supplement

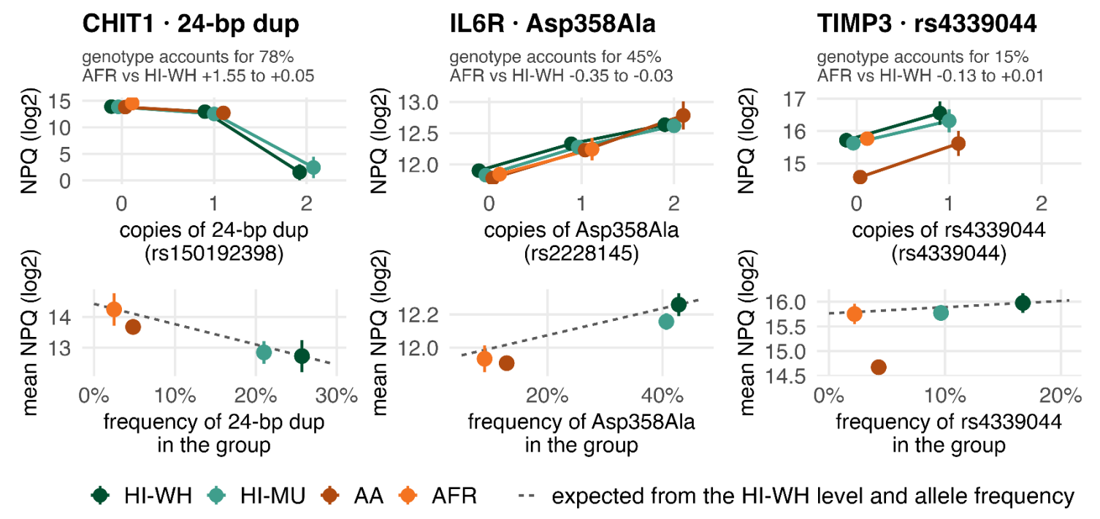
1. ANCESTRY MOVES THE BASELINE

All 49 African vs HI-WH differences point the same way in both lots



- 62 of 126 differ; vs HI-WH at NCI: African 49, AA 38, HI-MU 3
- ρ 0.93 between lots, size ratio 0.99; 19 above 2-fold; lots disagree for 5 (‡), reagent changed for 3 (†): lot 1 shown
- HI-MU, Peruvian and Ecuadorian vs Caribbean: 10 in one site, 22 across three (section 3)

The CHIT1 difference is the allele frequency; for IL6R, about half of it

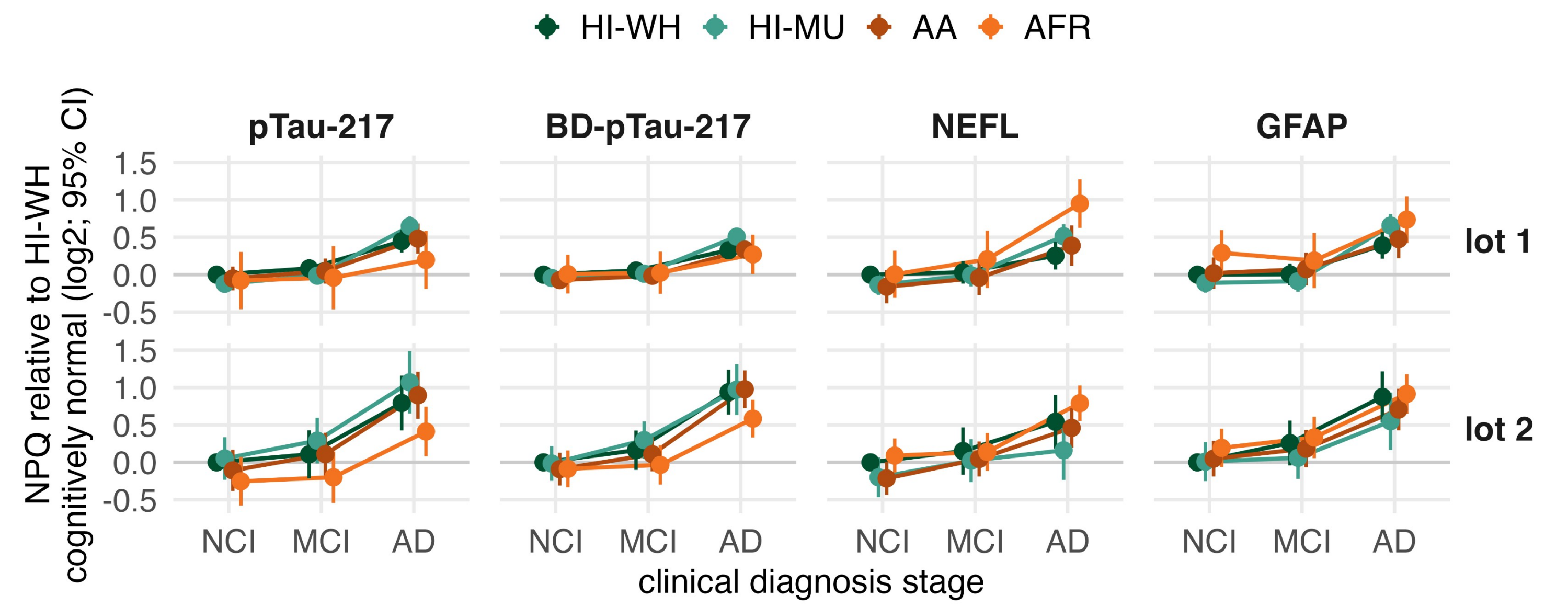


NPQ: NULISA Protein Quantification (log2), adjusted for covariates, lot, site and run.
Top: cognitively normal participants with WGS, by allele dose within group.
Bottom: each group's mean at NCI against its allele frequency; dashed = expected: the HI-WH level moved by 2 × per-allele effect × the frequency difference. % = share of the African vs HI-WH difference the confirmed variants predict (IL6R: with Duffy-null). Subtitle: African vs HI-WH before / after the genotype.
All African participants with WGS are in lot 2: their level rests on the lot adjustment.

Top: by allele dose within group, cognitively normal with WGS. Bottom: each group's mean NPQ at NCI against its frequency; dashed = expected from the HI-WH level and the per-allele effect; % = whole-cohort share. Before / after the genotype enters the model, sequenced participants, both lots in one fit with a lot term: CHIT1 +1.55 to +0.05 log2, IL6R −0.35 to −0.03, TIMP3 −0.13 to +0.01 (cohort −1.23). WGS 2,278 of 4,273; 30 of the 49 have no variant in the WGS set, 9 have one that does not move the protein.

2. ANCESTRY RARELY MOVES THE AD SIGNAL

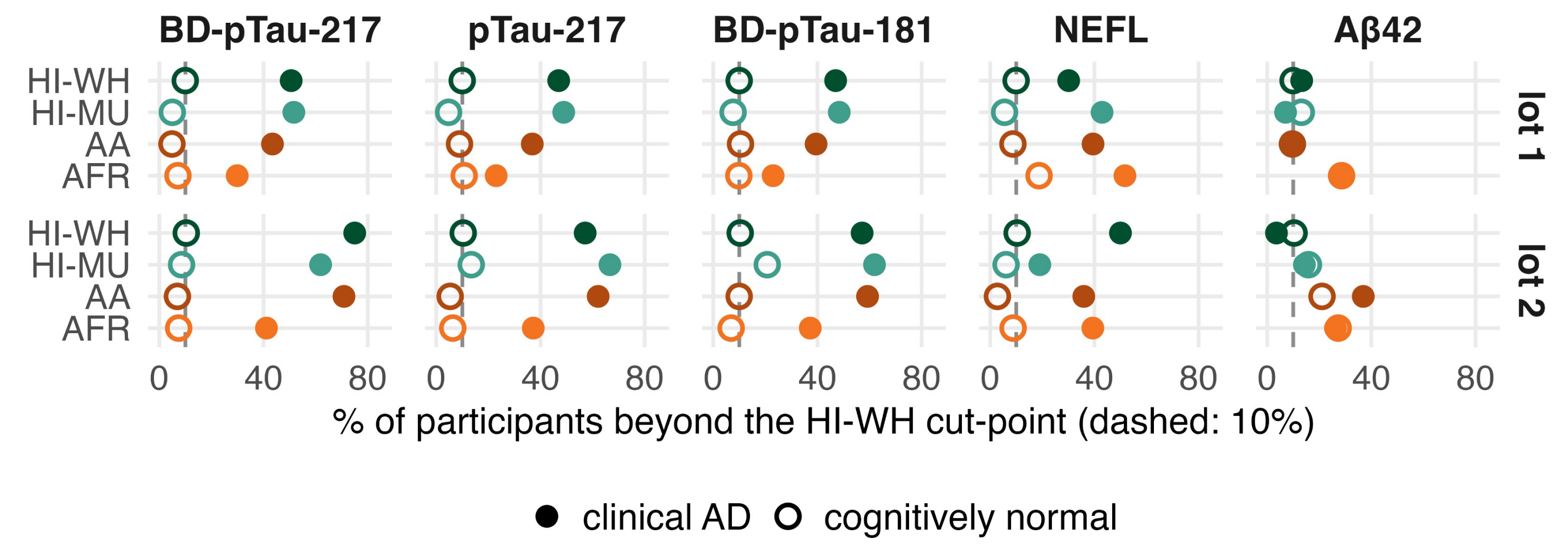
p-tau rises from NCI through MCI to AD in every group, in both lots; BD-pTau-217 rises least in African participants



Cross-sectional: different people at each stage, not one person followed over time.

- 12 group × diagnosis interactions of 126; BD-pTau-217 holds in lot 2 alone (p 0.002), pTau-217 and NEFL do not

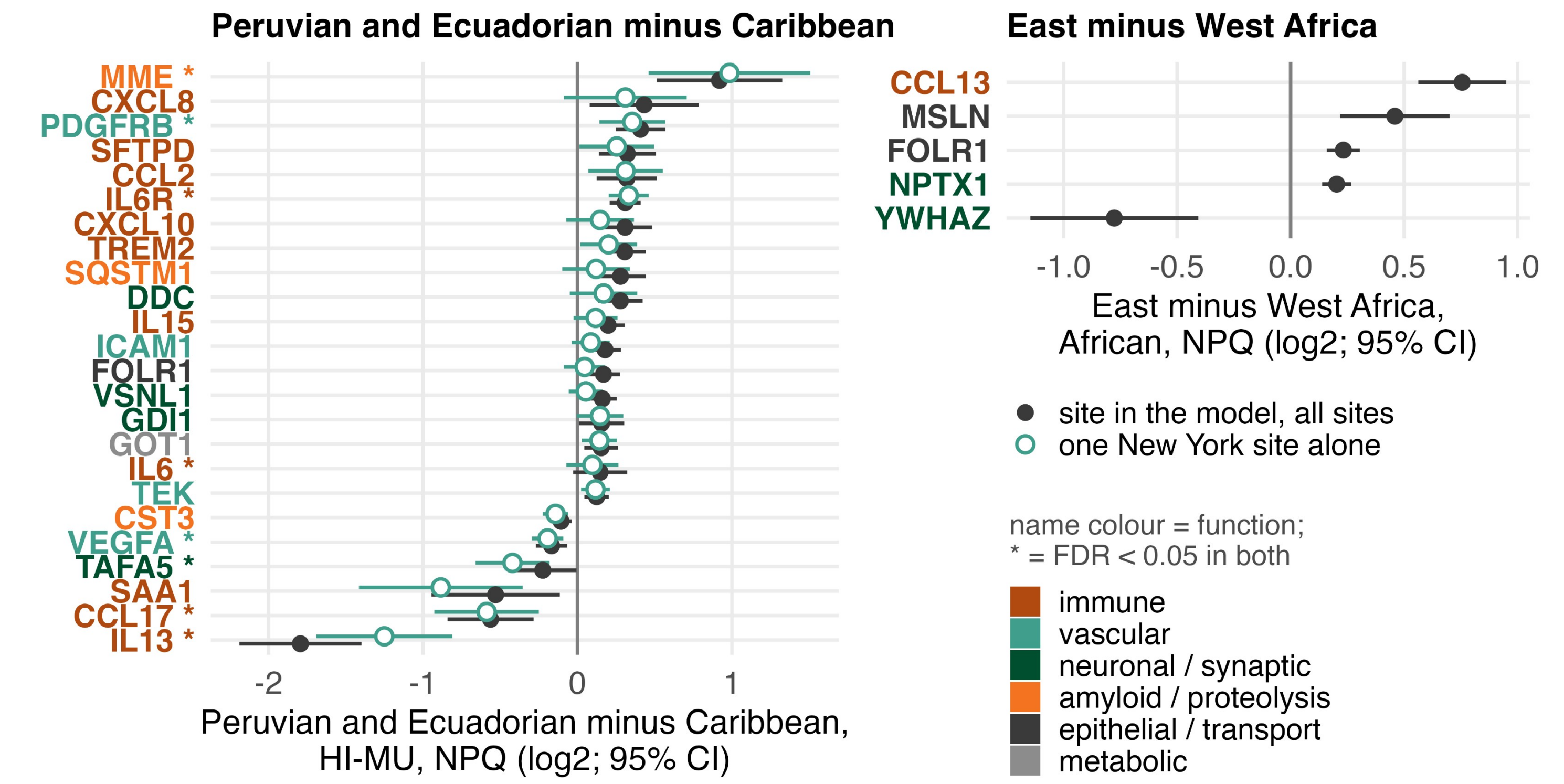
One BD-pTau-217 cut flags 5 to 10% of normals in every group, and fewer clinical AD at most African sites



Cut: 90th percentile of HI-WH cognitively normal, within lot (10th for Aβ42). BD-pTau-217, clinical AD above the cut: African 30% vs 43 to 52% (lot 1), 41% vs 62 to 75% (lot 2; HI-WH 28 AD / 58 NCI, HI-MU 21 / 82); by site in lot 2, 5 of 8 African sites 31 to 44%, 3 (50 to 75%) with the other groups' 50 to 76%; the same pattern on pTau-217 and BD-pTau-181.

3. REGIONS WITHIN A GROUP

Peruvian and Ecuadorian vs Caribbean: 10 differ inside one New York site; 22 across three sites, where the Peru site and Peruvian origin cannot be separated



Filled: all sites, site in the model (848 HI-MU); hollow: the New York site alone, 53 vs 379. The Peru site holds 197 of the 260 Peruvian and Ecuadorian and no Caribbean; * = FDR < 0.05 in both; the 14 that differ only across sites are all higher in Peruvian and Ecuadorian.

- East vs West Africa 5; 6 of the 10 + 5 are immune
- IL6R higher in Peruvian and Ecuadorian, as rs2228145 predicts

CAVEATS

- clinical diagnosis, not amyloid-confirmed; one draw per person
- groups: self-identified (US) or born in Africa; no genetic PCs
- African: own 13 sites, one assay run in lot 1
- handling, storage, diagnostic workup differ by site

SUPPLEMENT

detailed methods and the model, the model-term table, the ε4 figure, the six-arm interaction matrix, the five-panel cohort figure, per-lot tables, the full genotype table · github.com/emartin-um/ASHG2026-supplement

WHAT THIS MEANS

Normals cross the cut-point equally in every group; clinical AD does not.

p-tau: confirm AD with amyloid PET or CSF in every group before recalibrating the cut-point · plasma Aβ42 and Aβ42/40 do not separate AD in any group · a one-site population difference is a site difference until shown otherwise

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